MetaCore: Unique core sequences from metagenomic samples for microbiome wide association studies

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Objetive:

To extract the genetic information shared by multiple metagenome assemblies with a common phenotype and test its application in a microbiome wide association study (MWAS) in a case-control scenario

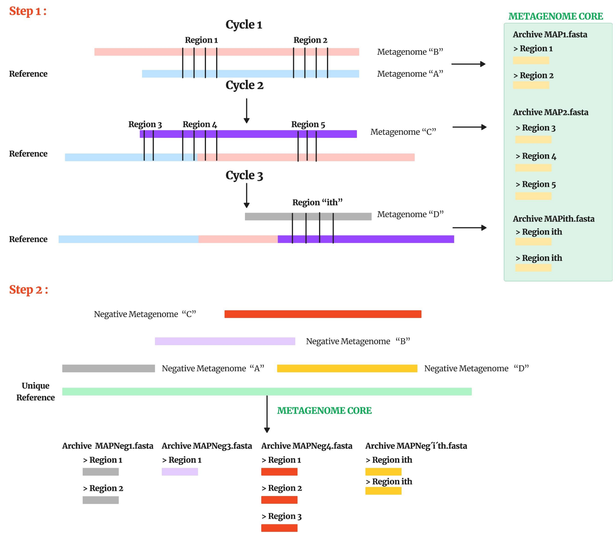
Introduction

Understanding the intricate relationship between genetic variations and observable traits has been the main challenge of genetic association studies. When applying association methods to microbial communities, the taxonomic, gene content, or diversity (alpha or beta) information has been tried to be linked to a particular human phenotype. However, little is known regarding environmental phenotypes, such as the degradation of textile dyes, from the eyes of association studies, as many taxonomical and genetically different bacteria can express under selective pressure such phenotype. An in-depth understanding of the genotype-phenotype association in microbial communities exposed to such compounds could lead to the development of better bioremediation strategies, as textile dyes molecules are a threat to the environment due to the great impact that these contaminants can have on ecosystems because of their recalcitrant nature and overall toxicity, with a high potential risk for human health. Here, we have developed a bioinformatic pipeline, called "MetaCore" (https://github.com/AyixonLab/MetaCore), that considers each microbial community as individual genomic entities within, but not limited to, a case-control genotype-phenotype association study with the goal of finding the common genetic information within phenotype-positive samples.

Methodology

Eighty-four metagenomic assemblies (23 Positive, 61 Negative) of environmental microbiomes available in the NCBI for the textile dye degradation phenotype were used in this study they comprised a total of 41 Gb of information. The MetaCore is obtaining by the used of MashMap which allows to estimates an unbiased k-mer based Jaccard similarity and identity estimates between contigs with a selected length (1) (**Step 1**) in each cycle a file containing the shared sequences with 85% of identity between positive samples is created. Next, all sub files are concatenated to create the MetaCore, which is used as reference (**Step 2**) to obtain the sequences that the negative samples shared with the MetaCore with the same parameters as step 1. A MWAS was perform at the genetic level using the metagenome assemblies of all samples and the sub files created by the MetaCore process. Taxonomic classification was performed with kraken2-bracken (2,3) using the RefSeq database (sept 2023), taxonomic assignments present in less than 20% of the samples, with less than 0.001% of reads and mOTUs with a relative abundance less than 0.1% were removed, the resulting matrix was used for estimating Bray-Courtis distance metric and ANOSIM was apply two assess the effect of the MetaCore to differentiate between groups. Additionally, MiRKAT is used to transform the Bray-Courtis matrix to a kinship matrix (4). Unitig-caller is used to call the genetic variants (unitigs) (5). Finally, the association of genetic variants within the microbiomes and the MetaCore with the phenotype is tested using a linear mixed model (Fast-LMM) implemented in Pyseer (6).

Illustration of the Obtention of Metagenomic Core Sequences (MetaCore)



Results and Discussion

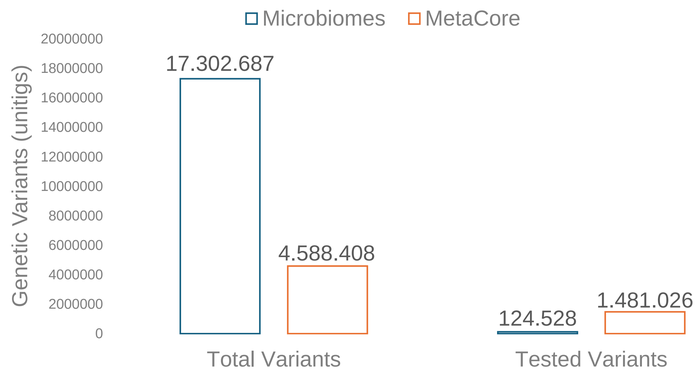


Figure A: Comparison of the number of genetic variants in the Textile Dye degradation MWAS

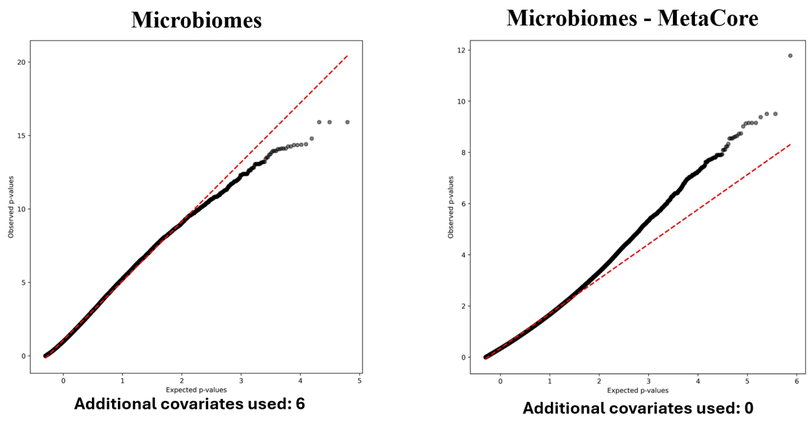


Figure B: Quantile-Quantile plot of the best association model result between the MWAS using the whole genomic information and the MetaCore

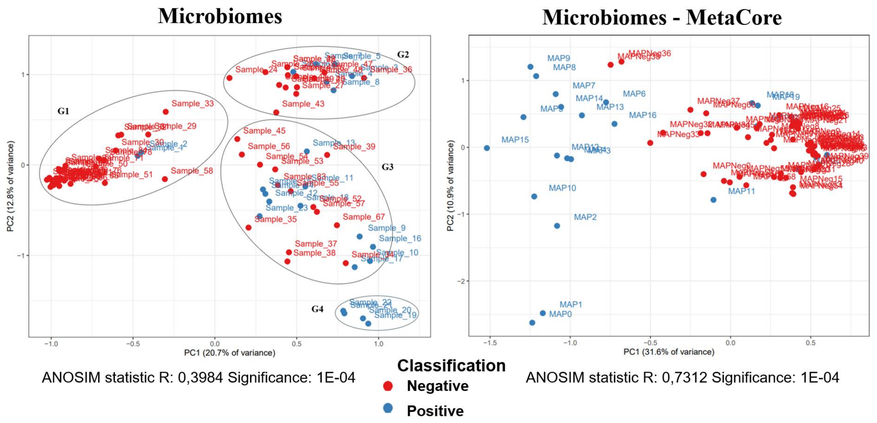


Figure C: Principal component analysis using the Bray-Courtis distance metric at the taxonomic genus level using the whole genomic (Microbiomes) information and the MetaCore. G1-G4

- One of the advantages conferred by the MetaCore is the reduction of the total amount of genomic information to be included in the association study. Though its application, the total number of genetic variants was reduced by 74% while maintaining space to find genetic associations (Fig. A). The use of the MetaCore allows including in absolute terms a greater number of genetic variants in the same frequency range (Fig. A, Tested variants). In this case, the best result when inspecting the quantile graphs (Fig. B) was obtained including the variants when frequency equal to or greater than 20%.
- Additionally, it is well-known that the inclusion of covariates in an association model is good practice among others to control for confounding variables as well as improves model fit, in our study was found that a good model fit is achieved with the MetaCore even if no covariates were used (Fig. B).
- It was also analyzed the effect of the MetaCore over the grouping of the samples based in its dissimilarities, It was found that the MetaCore achieved a strong separation of both groups (Fig. C) which also serves as an initial way to correlate the process of obtaining the MetaCore with the ability to select genetic information shared between samples with a common phenotype.
- Finally, the genes over which the significant genetic variants (p-value < 1E-08, Bonferroni corrected) mapped were explored allowing us to find that an isoform of the gene nouF [NADH-quinone oxidoreductase subunit F], the rpoC [DNA-directed RNA polymerase subunit beta] and a transposase of the IS1380 family were associated with the phenotype. Interestedly, the first two genes have taxonomic assignment related to *Thaurea* spp.

Conclusions

- It was possible through the use of unitigs to find reported actors in the textile dye degradation phenotype both at the gene level with the presence of nuoF and the taxonomic level with Thaurea spp. Additionally, it is interesting how throughout the study of this phenotype, actors with cryptic interpretation are revealed, such as the presence of transposases.
- **Full Metagenome** data increases the dimensionality and the expensiveness of computational resources, we propose the **MetaCore** alternative, which reduces the complexity as much as three times and increasing p-value astringency, which also helps to reduce false positive variants in the MWAS.
- An important aspect of the type of MWAS proposed in this work - unlike those carried out based on the relative abundance of genes or taxa, which use linear models test the association consisted of considering a matrix of relationships between the microbiomes -and in the MetaCore files- as fixed effects within the model to treat the Microbiome as an independent genomic entity.

Perspectives

- The inclusion of the greatest amount of information can lead to better modeling of the problem to provide more robust responses, to capture a larger population of unitigs not captured by the limited number of current samples. The above will allow testing genetic variants of a lower frequency (<20%).
- Performing an in-depth comparison of the MetaCore algorithm with a group of synthetic data will be an additional validation test necessary both to understand the limitations of computations and its applicability in other experimental designs.
- Differentiating true from false positive associations remains a challenge; our results should be taken cautiously, but it could be helpful to find a new way of analyzing environmental microbial phenotypes.

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